

Part 1: Data Analysis and Insight

1) Data overview:

Data source: The dataset was provided by the lecturer of the course *MIS 311: Introduction to Business Analytics* at the Eastern International University (EIU) for the completion of Individual Assignment #1.

Dataset dimensions:

Number of rows: 202 (excluding the header row)

Number of columns: 7

The context or background: the data set provides information about the used car market, including condition, various technical specifications, and pricing of vehicles. This dataset is highly valuable for car retailers and digital marketplaces to understand the price determinants, predict vehicle residual values, and optimize inventory management based on consumer demand and market trends.

Data column descriptions:

- Model (object): Name of the car
- Year (int): Year manufactured
- Km/h (int): Amount of distance moved
- Type (object): Automatic or manual car
- Fuel (object): Type of engine fuel
- Price (int): price of car

2) Data cleaning

To make sure data is accurate, it was cleaned by 2 main steps: removing rows with missing data (blanks) and removing duplicate rows.

Handling missing values: Using 'Go To Special > Blanks' tool in Excel to find empty cells. As a result, there were 10 rows with missing data; therefore, I used the 'Delete Entire Row' method to completely remove them.

Handling duplicate rows: Using 'Remove Duplicates' tool to check the repeated data. Excel found and removed 5 duplicate rows that were exactly the same

3) Descriptive statistics:

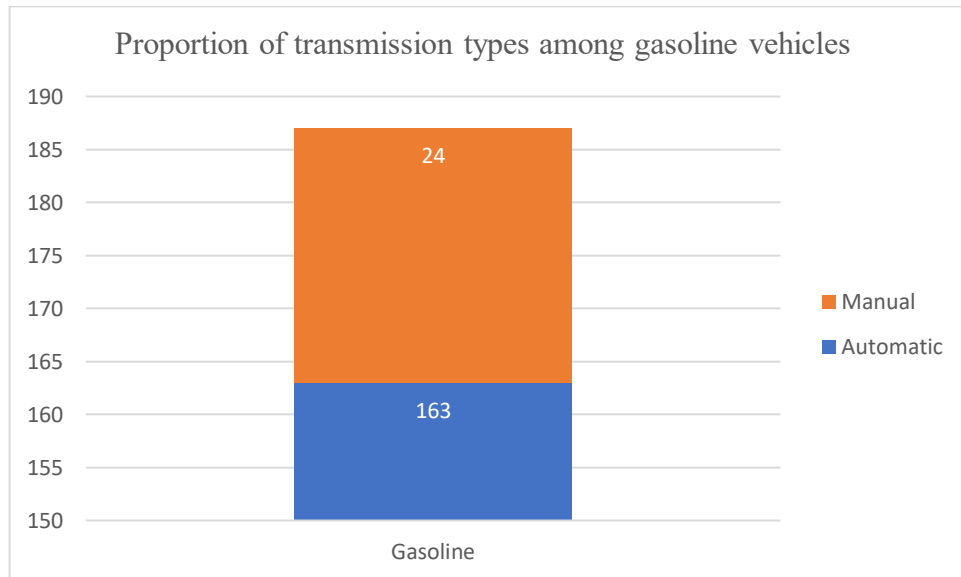
3.1 Numerical variables summary

Variable	Mean	Median	Min	Max	Stad. Deviation
Year	2016.45989	2017	2009	2024	4.91456536
Km/h	119158.369	102000	30	500000	78446.72507
Price	1037486.631	750000	220000	4800000	861553.9434

3.2 Categorical variables summary

Variable	Most Common Category	Frequency
Type	Automatic	163
Fuel	Gasoline	187
Color	Silver	44
Model	KIA Cerato	21

3.3 Visualisations and insights



Insight 1: Analysis shows that 100% of the cars are gasoline-powered, 87.2% are automatic transmission, and the most popular model is the KIA Cerato with 21 vehicles available.

Therefore, used car dealerships should focus their investment on models with these characteristics to sell them quickly.



Insight 2: The summary table shows that used car prices fluctuate significantly, ranging from a minimum of \$220,000 to a maximum of \$4,800,000, with an average price of \$1,037,487. This large price fluctuation is mainly due to the year of manufacture (from 2009 to 2024) and the mileage of the vehicle. Dealers can use this data to build accurate pricing and depreciation models based on age and mileage to establish competitive selling prices in the market.